AI-generated · Unofficial practice

Physics Mock

Paper 1 (Section A Multiple-choice · Section B Structured)

150 minutes（2½ hours）

FOR REVISION

For revision and self-assessment only. Not an official examination paper.

INSTRUCTIONS

1. This paper consists of Section A (multiple-choice) and Section B (structured questions).
2. Answer ALL questions.
3. Each question in Section A has only ONE correct answer and carries 1 mark.
4. In Section B, show all working clearly. Unless otherwise specified, give numerical answers to 3 significant figures.
5. A list of data, formulae and relationships is provided on page 2.
6. The diagrams in this paper are not necessarily drawn to scale.

Independent, unofficial mock paper. Not affiliated with any official examination authority.

Data, Formulae and Relationships

Data

acceleration due to gravity $g = 9.81\ \text{m s}^{-2}$ (close to the Earth)	universal gravitational constant $G = 6.67 \times 10^{-11}\ \text{N m}^2\ \text{kg}^{-2}$
speed of light in vacuum $c = 3.00 \times 10^{8}\ \text{m s}^{-1}$	charge of electron $e = 1.60 \times 10^{-19}\ \text{C}$
electron rest mass $m_e = 9.11 \times 10^{-31}\ \text{kg}$	permittivity of free space $\varepsilon_0 = 8.85 \times 10^{-12}\ \text{C}^2\ \text{N}^{-1}\ \text{m}^{-2}$
permeability of free space $\mu_0 = 4\pi \times 10^{-7}\ \text{H m}^{-1}$	atomic mass unit $u = 1.661 \times 10^{-27}\ \text{kg}$ （$1\ u \approx 931\ \text{MeV}$）
Avogadro constant $N_A = 6.02 \times 10^{23}\ \text{mol}^{-1}$	universal gas constant $R = 8.31\ \text{J mol}^{-1}\ \text{K}^{-1}$
Boltzmann constant $k = 1.38 \times 10^{-23}\ \text{J K}^{-1}$	Planck constant $h = 6.63 \times 10^{-34}\ \text{J s}$
Stefan constant $\sigma = 5.67 \times 10^{-8}\ \text{W m}^{-2}\ \text{K}^{-4}$	astronomical unit $1\ \text{AU} = 1.50 \times 10^{11}\ \text{m}$
light year $1\ \text{ly} = 9.46 \times 10^{15}\ \text{m}$	parsec $1\ \text{pc} = 3.09 \times 10^{16}\ \text{m}$

Motion

$v = u + at \qquad s = ut + \frac{1}{2}at^2 \qquad v^2 = u^2 + 2as \qquad s = \frac{1}{2}(u + v)t$

Force, energy, momentum

$F = ma \qquad F\Delta t = \Delta(mv) \qquad W = Fs\cos\theta \qquad E_P = mgh \qquad E_K = \frac{1}{2}mv^2 \qquad P = \frac{W}{t} = Fv$

Circular motion and gravitation

$a = \frac{v^2}{r} = \omega^2 r \qquad F = \frac{GMm}{r^2} \qquad v = \omega r$

Heat and gases

$Q = mc\Delta T \qquad Q = ml \qquad pV = nRT \qquad pV = \frac{1}{3}Nm\overline{c^2} \qquad E_K = \frac{3RT}{2N_A}$

Waves

$v = f\lambda \qquad n = \frac{\sin\theta_1}{\sin\theta_2} \qquad \sin c = \frac{1}{n} \qquad \Delta y = \frac{\lambda D}{a} \qquad d\sin\theta = n\lambda \qquad \frac{1}{u} + \frac{1}{v} = \frac{1}{f}$

Electricity and magnetism

$F = \frac{Q_1 Q_2}{4\pi\varepsilon_0 r^2} \qquad E = \frac{V}{d} \qquad V = IR \qquad R = \frac{\rho l}{A} \qquad P = IV = I^2R \qquad F = BIl\sin\theta \qquad F = BQv\sin\theta$

$B = \frac{\mu_0 I}{2\pi r} \qquad B = \frac{\mu_0 NI}{l} \qquad \varepsilon = N\frac{\Delta\Phi}{\Delta t} \qquad \frac{V_s}{V_p} = \frac{N_s}{N_p}$

Radioactivity and nuclear energy

$N = N_0 e^{-kt} \qquad A = kN \qquad t_{1/2} = \frac{\ln 2}{k} \qquad \Delta E = \Delta mc^2$

＊Original compilation of commonly used quantities; symbols follow usual physics conventions.

Section A Multiple-choice Questions

1. Hot water at $70\ °C$ is poured into a cup containing $0.40\ kg$ of water at $15\ °C$. The final temperature of the mixture is $30\ °C$. Assuming no energy is transferred to the cup or the surroundings, what is the mass of the hot water poured in?

A. $0.0857\ kg$ **B.** $0.109\ kg$

C. $0.150\ kg$ **D.** $0.171\ kg$

2. At $100\ °C$, boiling water changes into steam at the same temperature. Which of the following statements about the water molecules during this process is/are correct?

(1) The average kinetic energy of the molecules remains unchanged.

(2) The average separation between the molecules increases.

(3) The potential energy of the molecules decreases.

A. (1) only **B.** (2) only

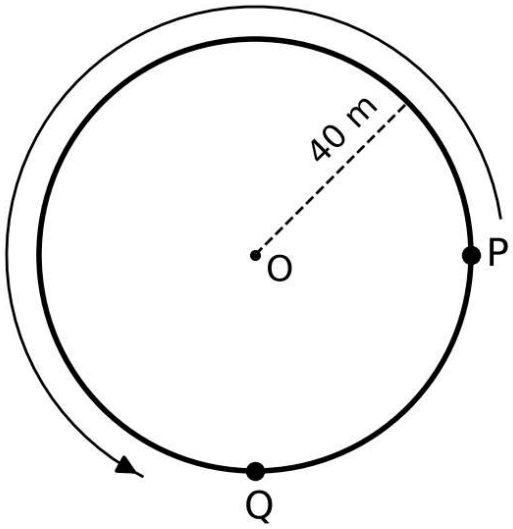
C. (1) and (2) only **D.** (1) and (3) only

3. A sealed steel tank of volume $0.050\ m^3$ contains $2.0\ kg$ of nitrogen at $17\ °C$. The molar mass of nitrogen is $0.028\ kg\ mol^{-1}$. Treating nitrogen as an ideal gas, what is the pressure in the tank?

A. $3.44 \times 10^3\ Pa$ **B.** $9.64 \times 10^4\ Pa$

C. $2.02 \times 10^5\ Pa$ **D.** $3.44 \times 10^6\ Pa$

4. A cyclist starts from P and rides anticlockwise for three-quarters of a lap along a circular track of radius **40 m** to Q, taking **30 s** (see the figure). What is the magnitude of the average velocity of the cyclist from P to Q?



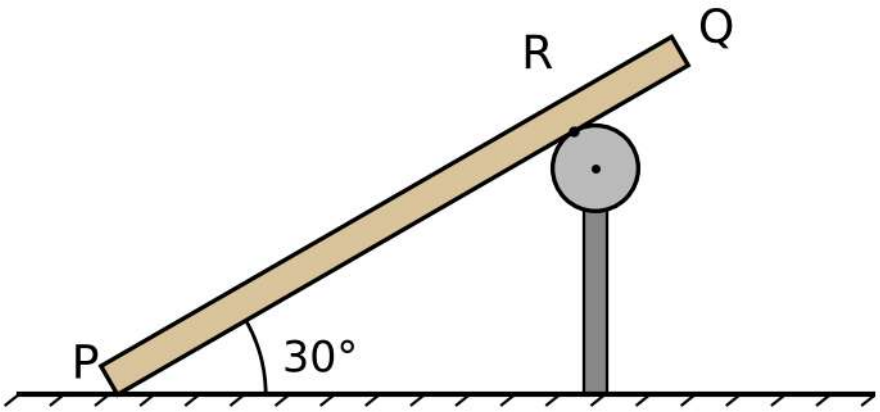
A. $1.33\ \mathrm{m\,s^{-1}}$

B. $1.89\ \mathrm{m\,s^{-1}}$

C. $2.67\ \mathrm{m\,s^{-1}}$

D. $6.28\ \mathrm{m\,s^{-1}}$

5. A uniform plank PQ of length $\mathbf{2.0\ m}$ and weight $\mathbf{40\ N}$ has its end P on rough horizontal ground and rests on a fixed smooth horizontal cylinder at R, where $\mathbf{PR = 1.6\ m}$. The plank makes $\mathbf{30°}$ with the horizontal and stays at rest (see the figure). What is the friction acting on end P by the ground?



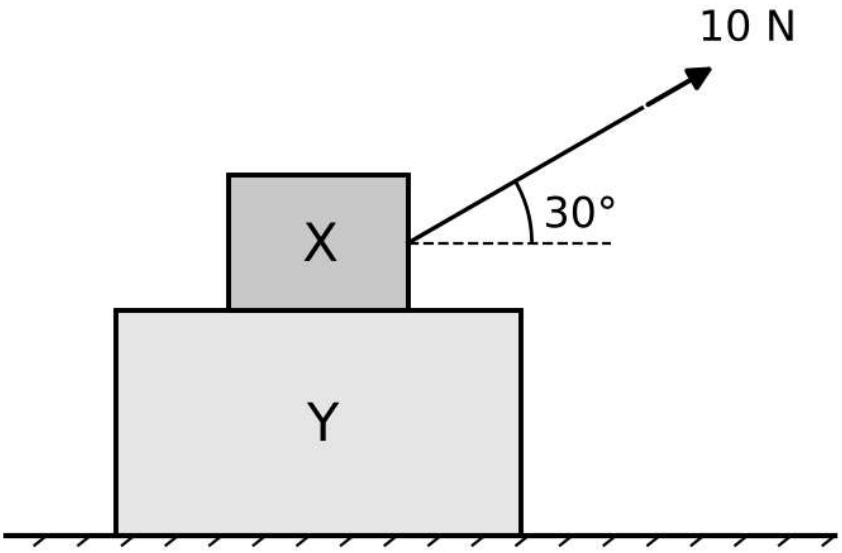
A. $\mathbf{10.8\ N}$, to the right

B. $\mathbf{10.8\ N}$, to the left

C. $\mathbf{18.8\ N}$, to the right

D. zero

6. Box X (mass **2.0 kg**) rests on box Y (mass **5.0 kg**), which stands on horizontal ground. A light string attached to X pulls it with a force of **10 N** at **30°** above the horizontal (see the figure). Both X and Y remain at rest. What is the normal reaction acting on Y by the ground?



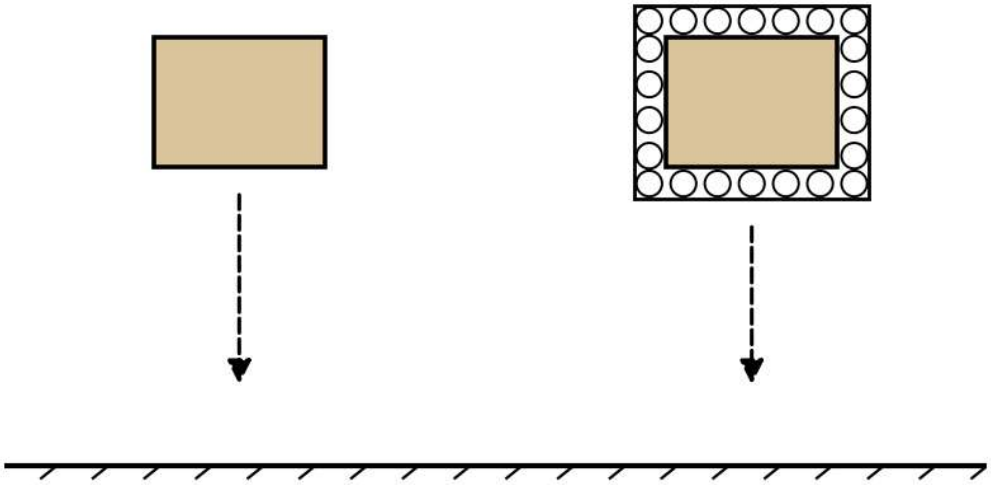
A. **63.7 N**

B. **68.7 N**

C. **73.7 N**

D. **78.7 N**

7. A parcel wrapped in bubble wrap is less likely to be damaged when it falls from a certain height onto hard ground than one without bubble wrap (see the figure). Which of the following is the best explanation?



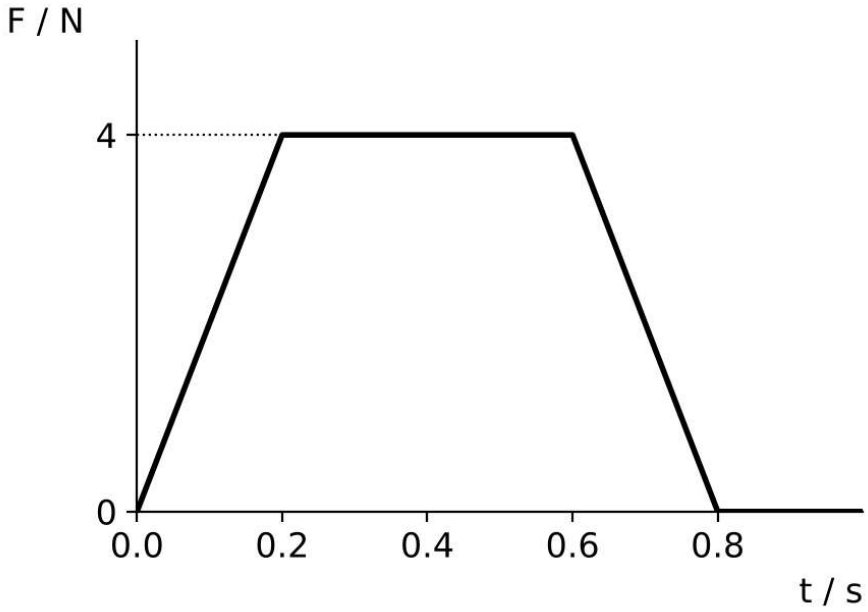
A. The bubble wrap reduces the change in momentum of the parcel.

B. The bubble wrap reduces the impulse on the parcel.

C. The bubble wrap reduces the gravitational force on the parcel.

D. The bubble wrap lengthens the time of impact, so the average impact force on the parcel is smaller.

8. A puck of mass $\mathbf{0.50\ kg}$ moves to the right at $\mathbf{3.0\ m\ s^{-1}}$ on smooth ice. From $\boldsymbol{t = 0}$, a force $\boldsymbol{F}$ directed to the left acts on the puck. The variation of $\boldsymbol{F}$ with time $\boldsymbol{t}$ is shown in the figure. What is the velocity of the puck at $\boldsymbol{t = 0.8\ s}$?



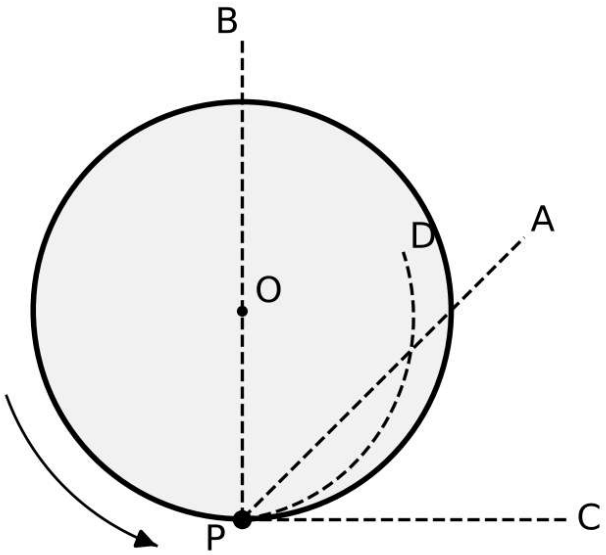
A. $\mathbf{0.20\ m\ s^{-1}}$, to the left

B. $\mathbf{1.8\ m\ s^{-1}}$, to the left

C. $\mathbf{4.8\ m\ s^{-1}}$, to the left

D. $\mathbf{7.8\ m\ s^{-1}}$, to the right

9. A horizontal circular platform rotates uniformly about its centre O, anticlockwise as seen from above. A girl standing at point P on the edge of the platform throws a ball horizontally towards O relative to herself. As seen from above, which path in the figure is the ball most likely to follow relative to the ground after release? (Neglect air resistance.)



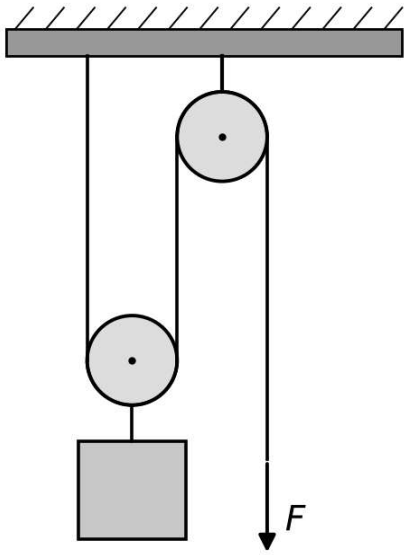
A. Path A

B. Path B

C. Path C

D. Path D

10. In the pulley system shown, a load of **400 N** hangs from a movable pulley of weight **40 N**. A worker pulls the free end of the rope vertically downwards so that the load rises at constant speed. The free end is pulled down by **3.0 m** and the worker does **800 J** of work. How much energy is used to overcome friction in this process? (The rope is light and inextensible.)



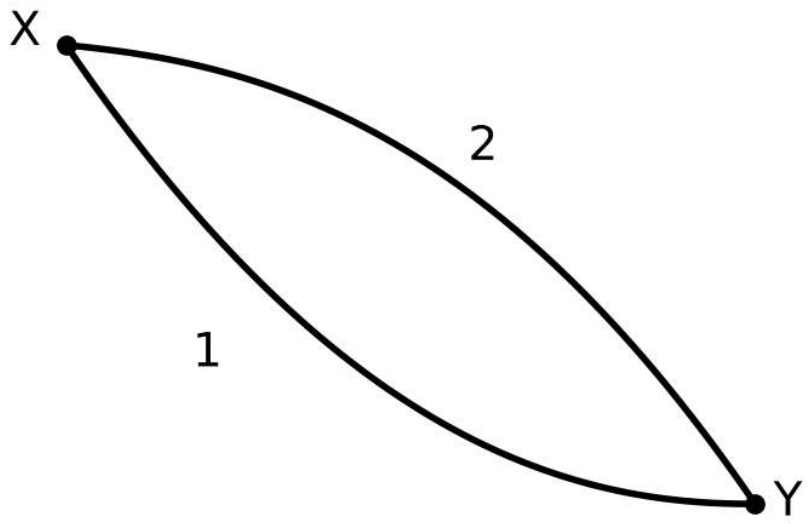
A. 140 J

B. 200 J

C. 360 J

D. 470 J

11. Two smooth slides, 1 and 2, both run from point X to point Y and have the same length (see the figure). Two identical small objects are released from rest at X at the same time, one on each slide. Which of the following is correct?



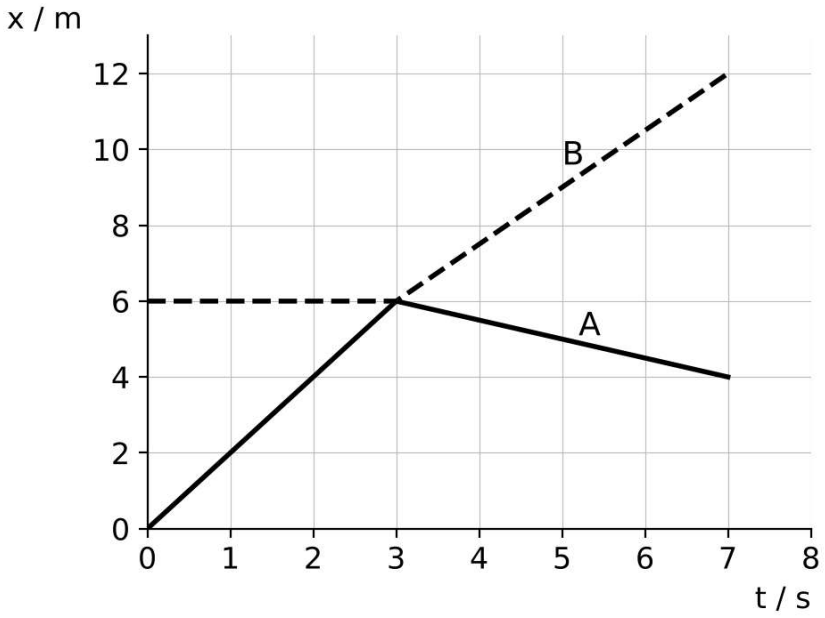
A. They reach Y with the same speed and at the same time.

B. The object on slide 1 reaches Y with a greater speed and arrives earlier.

C. They reach Y with the same speed, and the object on slide 2 arrives earlier.

D. They reach Y with the same speed, and the object on slide 1 arrives earlier.

12. On a smooth horizontal track, trolley A of mass **1.0 kg** collides with a stationary trolley B. The figure shows the variation of the positions x of the two trolleys with time t. What is the mass of B, and is the collision elastic?



A. **1.0 kg**, inelastic

B. **1.7 kg**, inelastic

C. **2.5 kg**, elastic

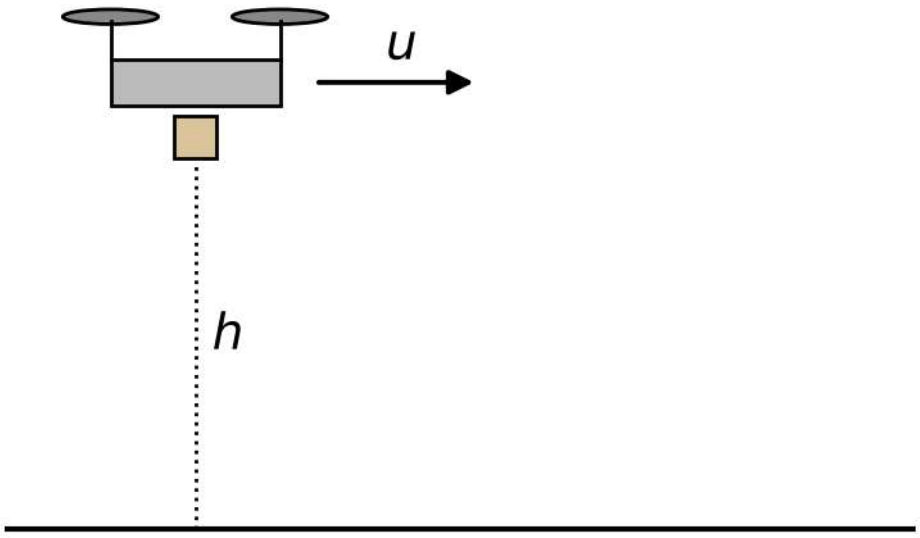
D. **1.7 kg**, elastic

13. A drone flies horizontally at speed u at a height h above the ground and releases a small parcel (see the figure). If the drone instead flies horizontally at speed $2u$ at the same height and releases the same parcel, which of the following is/are correct? (Neglect air resistance; the drone keeps flying at constant velocity.)

(1) In both cases, the drone is directly above the parcel when the parcel lands.

(2) The horizontal displacement of the parcel from release to landing is doubled.

(3) The speed of the parcel on hitting the ground is doubled.



A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

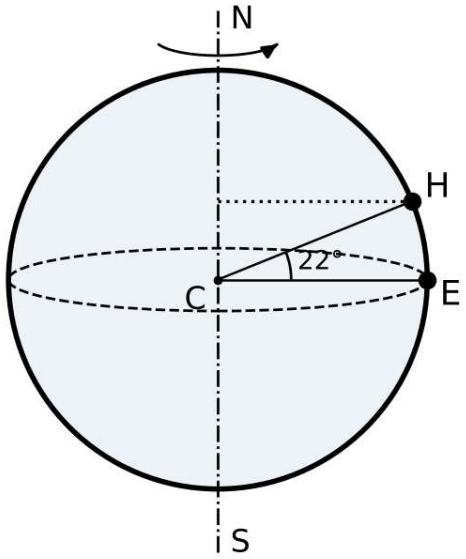
D. (1), (2) and (3)

14. The figure shows the rotating Earth. E is a point on the equator and H is a point at latitude **22°** N. Which of the following is/are correct?

(1) E and H have the same angular velocity about the Earth's axis.

(2) The linear speed of E is greater than that of H.

(3) The centripetal acceleration of H is directed towards the centre of the Earth.



A. (1) only

B. (1) and (2) only

C. (1) and (3) only

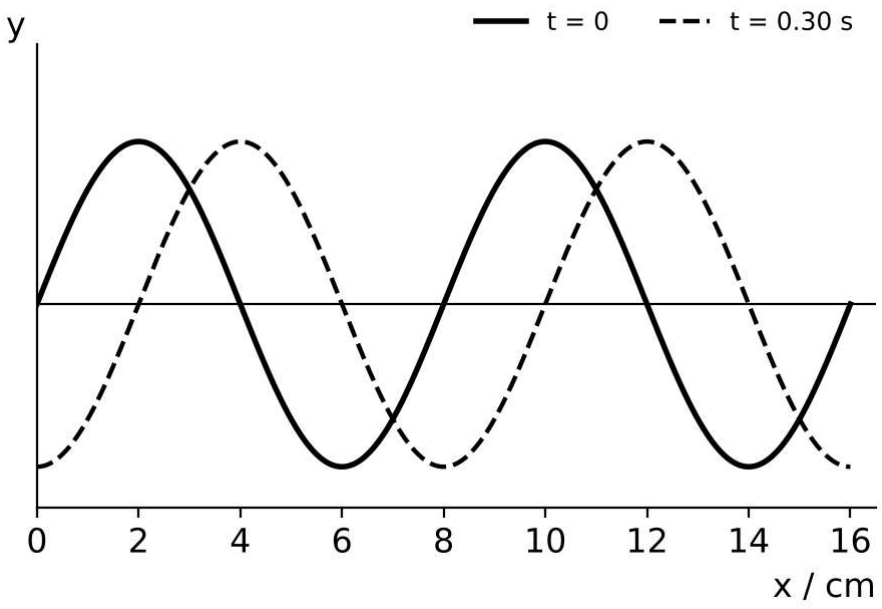
D. (1), (2) and (3)

15. A transverse wave travels in the positive x-direction. The solid and dashed curves in the figure show the waveform at $t = 0$ and $t = 0.30\ \text{s}$ respectively. Which of the following is/are correct?

(1) The particles whose equilibrium positions are at $x = 2\ \text{cm}$ and $x = 6\ \text{cm}$ always move in opposite directions.

(2) The period of the wave could be $0.24\ \text{s}$.

(3) At $t = 0$, the particle whose equilibrium position is at $x = 0$ is moving upwards.



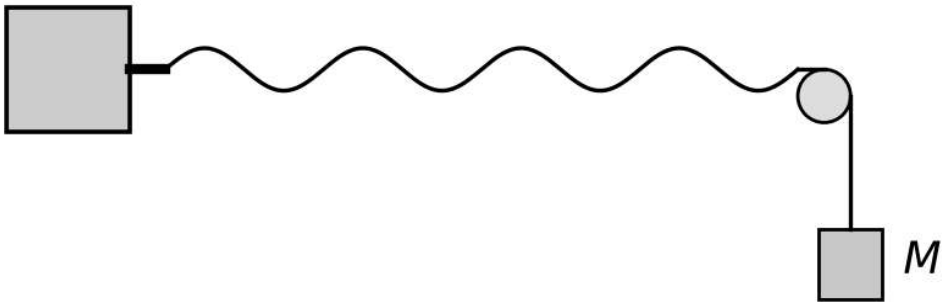
A. (1) only

B. (2) only

C. (1) and (2) only

D. (1), (2) and (3)

16. One end of a string is attached to a vibrator of fixed frequency; the other end passes over a pulley and carries a hanging mass M. The transverse waves travelling on the string have wavelength λ (see the figure). If the hanging mass is replaced by one of mass $4M$ and the frequency of the vibrator is unchanged, what is the new wavelength of the transverse waves on the string?



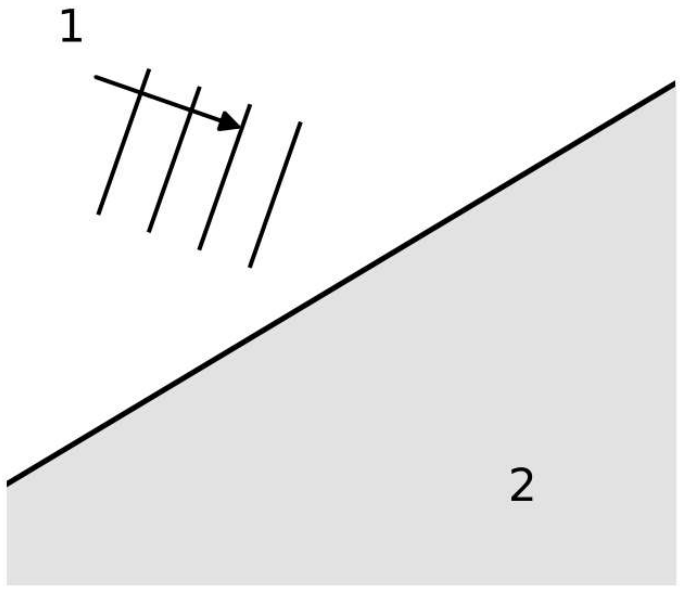
A. $\frac{\lambda}{2}$

B. λ

C. 2λ

D. 4λ

17. Water waves pass obliquely from deep region 1 into shallow region 2 (see the figure). After entering region 2, how do the wavelength and frequency of the waves change, and how is the direction of travel bent?



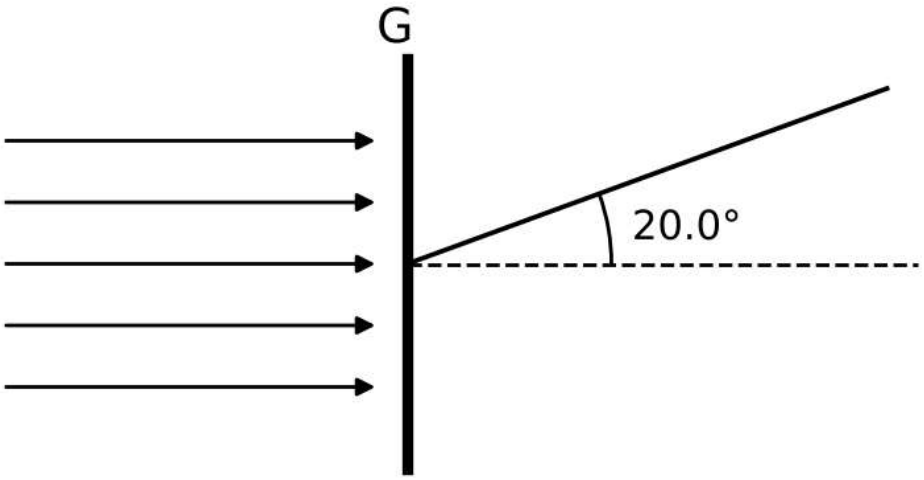
A. wavelength decreases; frequency decreases; bends towards the normal

B. wavelength decreases; frequency unchanged; bends towards the normal

C. wavelength increases; frequency unchanged; bends away from the normal

D. wavelength unchanged; frequency decreases; bends towards the normal

18. A beam of monochromatic light is incident normally on a diffraction grating. The first-order bright fringe is at an angle of $20.0°$ (see the figure). What is the angle of the second-order bright fringe, and what is the maximum number of bright fringes (including the central one) that can be observed?



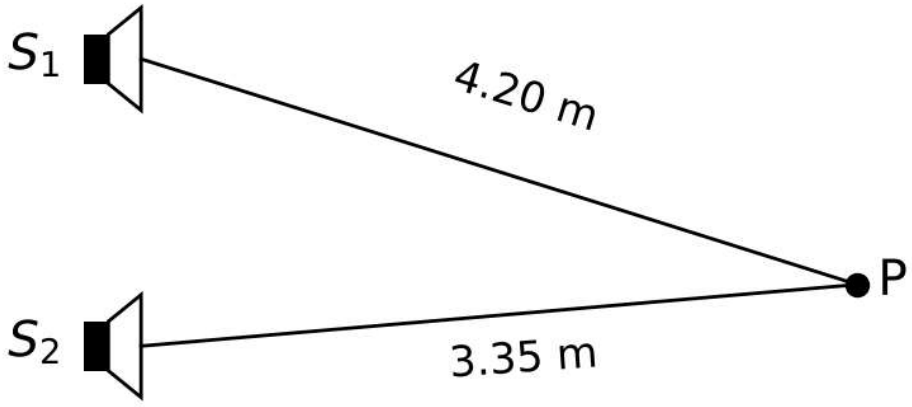
A. $43.2°$; 5

B. $43.2°$; 6

C. $40.0°$; 5

D. $40.0°$; 7

19. Two loudspeakers S_1 and S_2 are driven by the same signal generator but vibrate in antiphase. Point P is $4.20\ \text{m}$ from S_1 and $3.35\ \text{m}$ from S_2 (see the figure). The speed of sound is $340\ \text{m s}^{-1}$. Which frequency gives destructive interference at P?



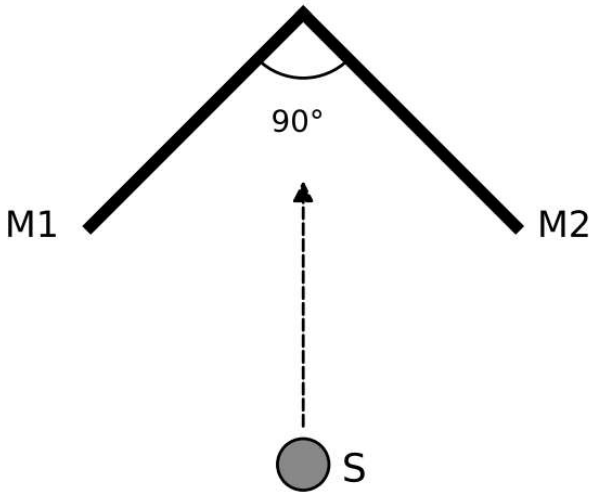
A. $200\ \text{Hz}$

B. $300\ \text{Hz}$

C. $600\ \text{Hz}$

D. $800\ \text{Hz}$

20. Two vertical plane mirrors are joined at $90°$. A student stands in front of them facing the joint and raises her right hand (see the top view in the figure). She looks at the image formed by one reflection in each mirror. Which of the following correctly describes this image?



A. The image appears to raise its left hand and is upright.

B. The image appears to raise its left hand and is inverted.

C. The image appears to raise its right hand and is inverted.

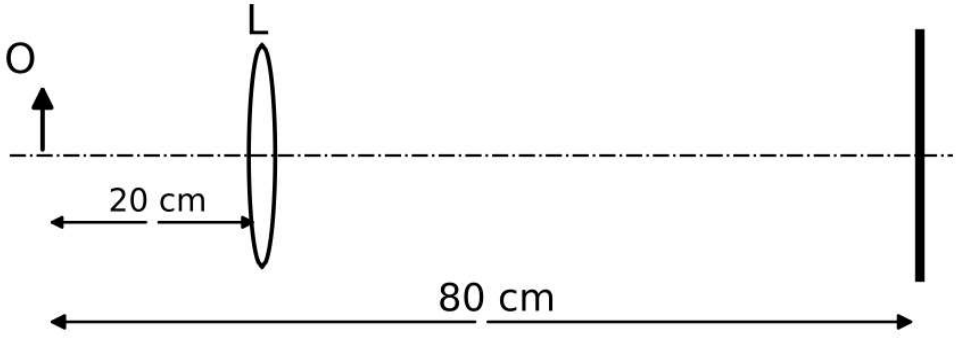
D. The image appears to raise its right hand and is upright.

21. An object and a screen are fixed $80\ \text{cm}$ apart. When a convex lens is placed between them $20\ \text{cm}$ from the object, a sharp image is formed on the screen (see the figure). Which of the following is/are correct?

(1) If the lens is moved to $60\ \text{cm}$ from the object, a sharp image is formed on the screen again.

(2) The focal length of the lens is $15\ \text{cm}$.

(3) With the lens at the second position, the image on the screen is larger than the object.



A. (1) and (3) only

B. (1) and (2) only

C. (2) and (3) only

D. (1), (2) and (3)

22. Which of the following statements about ultrasound is/are correct?

(1) Its frequency is higher than $\mathbf{20\ kHz}$.

(2) It travels faster in air than in water.

(3) It can be reflected and diffracted.

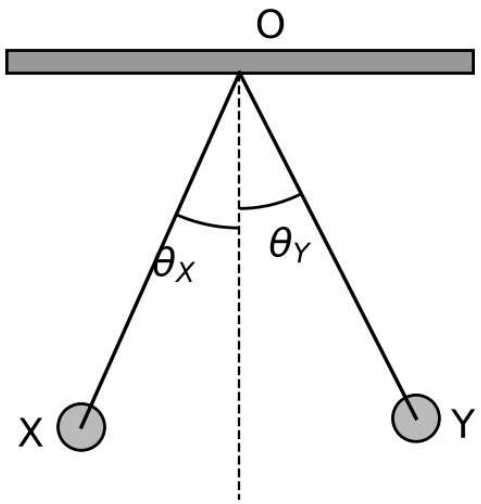
A. (1) only

B. (1) and (3) only

C. (2) and (3) only

D. (1), (2) and (3)

23. Two small charged spheres X and Y hang from the same point O by insulating threads of equal length. X has mass $\boldsymbol{2m}$ and charge $\boldsymbol{+q}$; Y has mass $\boldsymbol{m}$ and charge $\boldsymbol{+3q}$. In equilibrium the threads make angles $\boldsymbol{\theta_X}$ and $\boldsymbol{\theta_Y}$ with the vertical (see the figure, not drawn to scale). Which of the following is correct?



未按比例 Not to scale

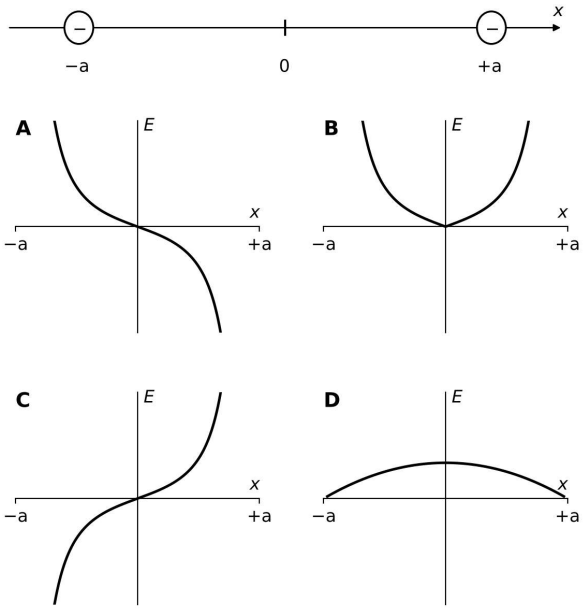
A. $\boldsymbol{\theta_X > \theta_Y}$

B. $\boldsymbol{\theta_X = \theta_Y}$

C. $\boldsymbol{\theta_X < \theta_Y}$, and $\boldsymbol{\sin\theta_Y = 6\sin\theta_X}$

D. $\boldsymbol{\theta_X < \theta_Y}$, and $\boldsymbol{\sin\theta_Y = 2\sin\theta_X}$

24. Two identical negative point charges $-Q$ are fixed at $x = -a$ and $x = +a$. Taking the $+x$ direction as positive, which graph best shows the variation of the electric field E along the x-axis between the charges ($-a < x < +a$)?



A. Graph A

B. Graph B

C. Graph C

D. Graph D

25. A mobile phone is charged with a steady current of $1.5\ \text{A}$ for 40 minutes. How many electrons pass through any cross-section of the charging cable during this time?

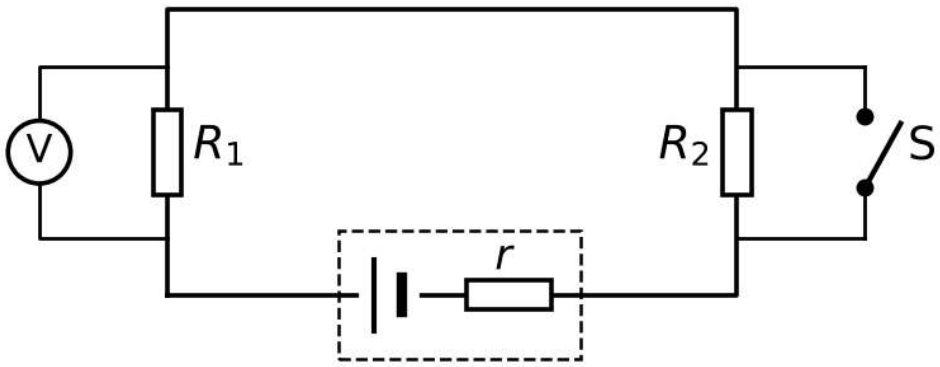
A. 9.38×10^{18}

B. 3.75×10^{20}

C. 2.25×10^{22}

D. 1.35×10^{24}

26. In the circuit shown, the battery has e.m.f. $\mathbf{12\ V}$ and internal resistance $\mathbf{2.0\ \Omega}$; $\boldsymbol{R_1 = 4.0\ \Omega}$ and $\boldsymbol{R_2 = 2.0\ \Omega}$. Switch S is connected in parallel with $\boldsymbol{R_2}$, and an ideal voltmeter V is connected across $\boldsymbol{R_1}$. What are the readings of the voltmeter when S is open and when S is closed?



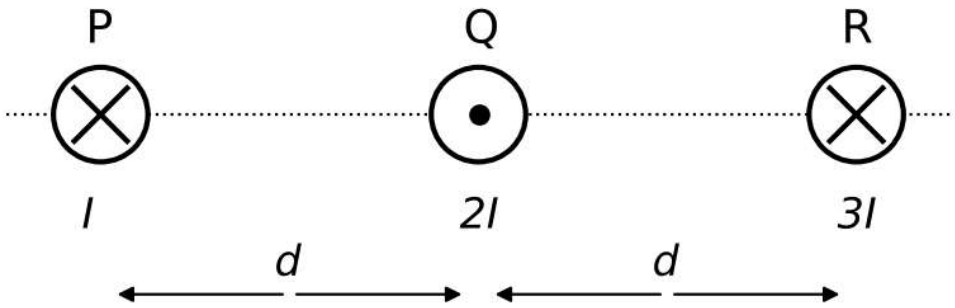
A. $\mathbf{6.0\ V; 8.0\ V}$

B. $\mathbf{8.0\ V; 12\ V}$

C. $\mathbf{6.0\ V; 12\ V}$

D. $\mathbf{8.0\ V; 8.0\ V}$

27. Three long straight parallel wires P, Q and R are perpendicular to the paper and equally spaced along a straight line, with adjacent wires a distance $\boldsymbol{d}$ apart. P carries a current $\boldsymbol{I}$ into the paper, Q carries $\boldsymbol{2I}$ out of the paper and R carries $\boldsymbol{3I}$ into the paper (see the figure). Let $\boldsymbol{F_0}$ be the force per unit length between two long straight wires a distance $\boldsymbol{d}$ apart, each carrying a current $\boldsymbol{I}$. What is the resultant force per unit length on R?



A. $\mathbf{3.0}\boldsymbol{F_0}$, away from P

B. $\mathbf{4.5}\boldsymbol{F_0}$, towards P

C. $\mathbf{4.5}\boldsymbol{F_0}$, away from P

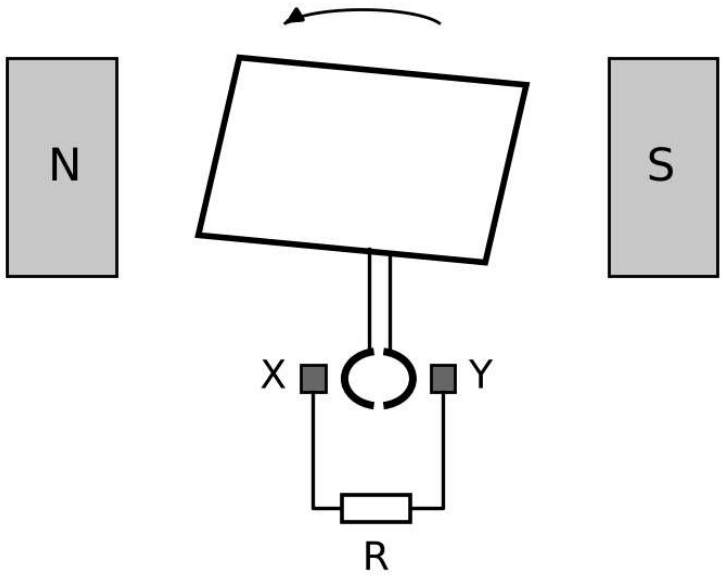
D. $\mathbf{7.5}\boldsymbol{F_0}$, away from P

28. The figure shows a simple d.c. generator. The coil rotates uniformly in a magnetic field and is connected to a resistor R through a commutator and brushes X and Y. Which of the following is/are correct?

(1) The commutator reverses the connections between the coil and the external circuit every half-turn, so the current through R is always in the same direction.

(2) If the rotation speed of the coil is doubled, both the peak output e.m.f. and the frequency of its pulses are doubled.

(3) The output e.m.f. is maximum when the plane of the coil is perpendicular to the magnetic field.



A. (1) only

B. (1) and (2) only

C. (1) and (3) only

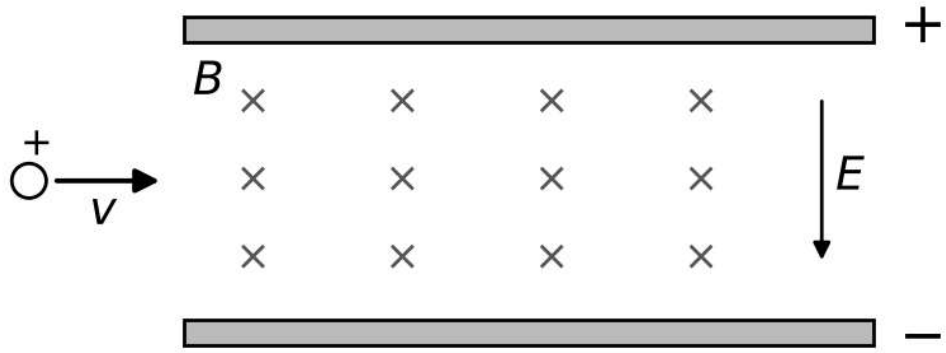
D. (1), (2) and (3)

29. Between a pair of parallel metal plates there is a uniform electric field $\boldsymbol{E}$ pointing downwards and a uniform magnetic field $\boldsymbol{B}$ pointing into the paper. A positively charged particle enters moving to the right with speed $\boldsymbol{v}$ and passes straight through (see the figure). Neglect gravity. Which of the following is/are correct?

(1) If only the magnetic field is reversed, the particle will be deflected towards the negative plate.

(2) A positively charged particle with speed $\boldsymbol{2v}$ will be deflected towards the positive plate.

(3) If $\boldsymbol{E}$ is doubled, $\boldsymbol{B}$ must be halved for the particle with speed $\boldsymbol{v}$ to pass straight through.



A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

D. (1), (2) and (3)

30. A lamp marked "**6.0 V**, **3.0 W**" is connected to a sinusoidal a.c. supply of peak voltage **6.0 V**. Assuming the resistance of the filament is constant, what is the average power consumed by the lamp?

A. **0.75 W**

B. **1.5 W**

C. **2.1 W**

D. **3.0 W**

31. A beam of X-rays has a wavelength of **0.050 nm**. Which of the following statements about this beam is correct?

A. They are deflected in a magnetic field.

B. Their wavelength is longer than that of ultraviolet radiation.

C. The energy of each photon is about **25 keV**.

D. The energy of each photon is about **25 eV**.

32. A sample contains two radioactive nuclides P and Q with half-lives of **4 h** and **12 h** respectively. At $t = 0$ the total activity of the sample is **800 Bq**; at $t = 12\ \text{h}$ it is **160 Bq**. What is the activity of P at $t = 0$?

A. **160 Bq**

B. **400 Bq**

C. **480 Bq**

D. **640 Bq**

33. The following is a nuclear fission reaction:

$$^{239}_{94}\mathrm{Pu} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{134}_{54}\mathrm{Xe} + {}^{A}_{40}\mathrm{Zr} + 3\,{}^{1}_{0}\mathrm{n}$$

What is the value of A?

A. 102

B. 103

C. 105

D. 106

END OF SECTION A

Section B Structured Questions

1. An alcohol thermometer without a scale has an alcohol column of length **3.2 cm** when placed in melting ice and **18.4 cm** when placed in boiling water. A student assumes that the boiling water is at **100 °C** and then puts the thermometer into a cup of warm water, where the column length is **11.0 cm**.

(a)(i) Find the temperature of the warm water. (2)

(ii) State one assumption made in the calculation in (a)(i). (1)

(b) It is later found that the boiling water used in the calibration was boiled at high altitude and was only at **98.0 °C**. Find the actual temperature of the warm water, and state whether the result in (a)(i) is an overestimate or an underestimate. (2)

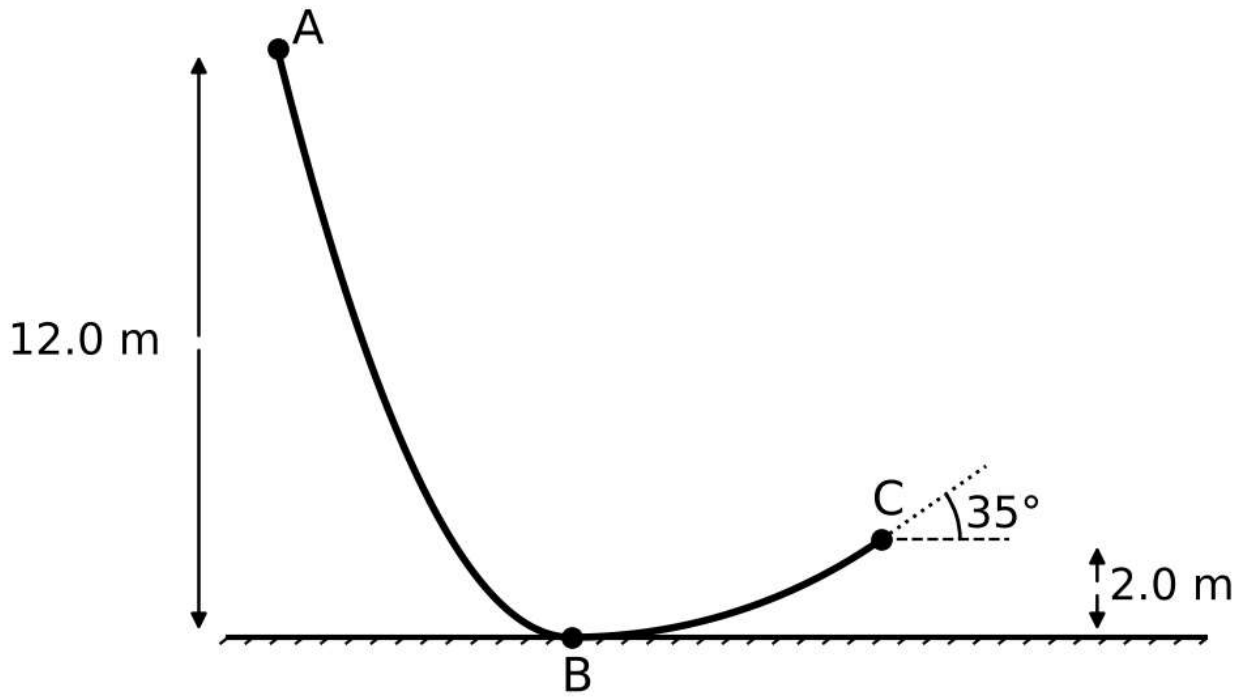
2. A party helium cylinder has an internal volume of $0.012\ m^3$ and contains helium at a pressure of $1.5 \times 10^7\ Pa$. Each balloon, when filled, has a volume of $0.0080\ m^3$ and contains helium at $1.2 \times 10^5\ Pa$. Assume the temperature of the helium stays constant throughout, and treat helium as an ideal gas.

(a) Find the volume that all the helium in the cylinder would occupy at a pressure of $1.2 \times 10^5\ Pa$. (2)

(b)(i) Helium can no longer flow into a balloon once the cylinder pressure falls to $1.2 \times 10^5\ Pa$. Find the volume of helium (measured at $1.2 \times 10^5\ Pa$) available for filling balloons. (1)

(ii) A party needs 200 balloons. Determine whether one cylinder of helium is enough. Support your answer with calculation. (3)

3. A skier starts from rest at A and slides down a smooth slope through the lowest point B, then up a ramp, leaving it at C where the direction of motion is **35°** above the horizontal. A is **12.0 m** above B and C is **2.0 m** above B; the horizontal level of B is also the landing ground (see Figure 1.3). Neglect air resistance and all friction, and treat the skier as a particle.



(a) Describe the energy conversion of the skier from A to C, and find the speed of the skier at C. (3)

(b)(i) Find the maximum height of the skier above the ground while in the air. (2)

(ii) Find the horizontal distance between C and the point where the skier lands. (3)

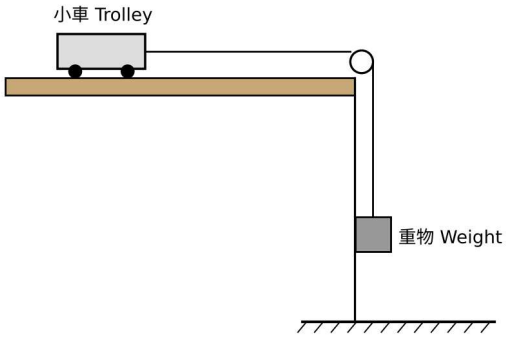
(c) A classmate says, "If the end of the ramp is made steeper (with the height of C unchanged), the skier will land with a greater speed." Comment on this statement. (1)

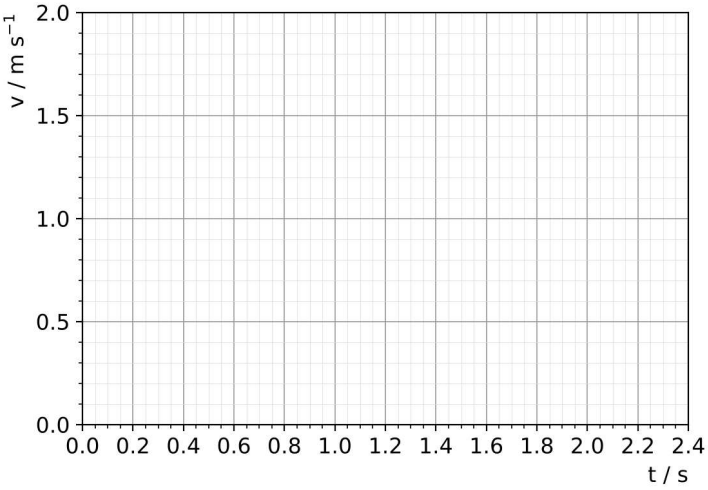
4. A trolley of mass **1.20 kg** is placed on a rough horizontal track and connected by a light string passing over a pulley to a hanging weight (see Figure 1.4, left). When released from rest, the trolley is pulled along; at some instant the weight hits the floor, after which the trolley continues to move. A motion sensor records the velocity v of the trolley as follows:

t / s: **0**, **0.2**, **0.4**, **0.6**, **0.8**, **1.0**, **1.2**, **1.4**, **1.6**, **1.8**, **2.0**

v / $\mathrm{m\,s^{-1}}$: **0.00**, **0.30**, **0.61**, **0.91**, **1.21**, **1.52**, **1.27**, **1.02**, **0.77**, **0.52**, **0.27**

Assume the friction on the trolley has a constant magnitude, the pulley is smooth and the string is inextensible.





(a) Draw a free-body diagram of the trolley during the first **1.0 s**, labelling all the forces acting on it. (2)

(b)(i) On the grid on the right of Figure 1.4, plot the v–t graph of the trolley. (2)

(ii) From the graph, find the magnitudes of the acceleration of the trolley during the first **1.0 s** and afterwards. (2)

(c) Explain why the trolley decelerates after $t = 1.0\ \mathbf{s}$. (2)

(d) From the results of (b)(ii), find the friction f on the trolley and the mass m of the hanging weight. (3)

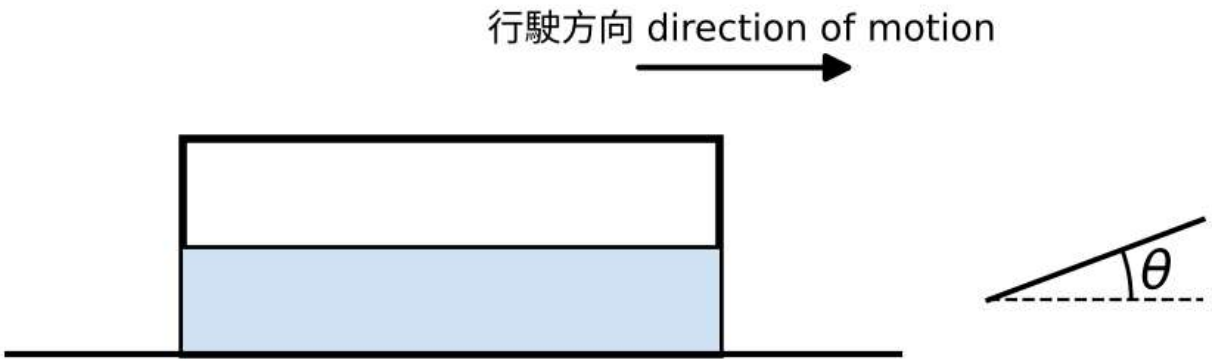
5. A student wants to measure the acceleration of a bus as it pulls away from a stop on a straight level road (assume the acceleration is roughly uniform). She has the following apparatus:

• a transparent sealed rectangular plastic box (length **30 cm**), half filled with coloured water

• a metre rule

• double-sided tape

When the bus accelerates, the water surface in the box tilts at an angle θ to the horizontal (see Figure 1.5).

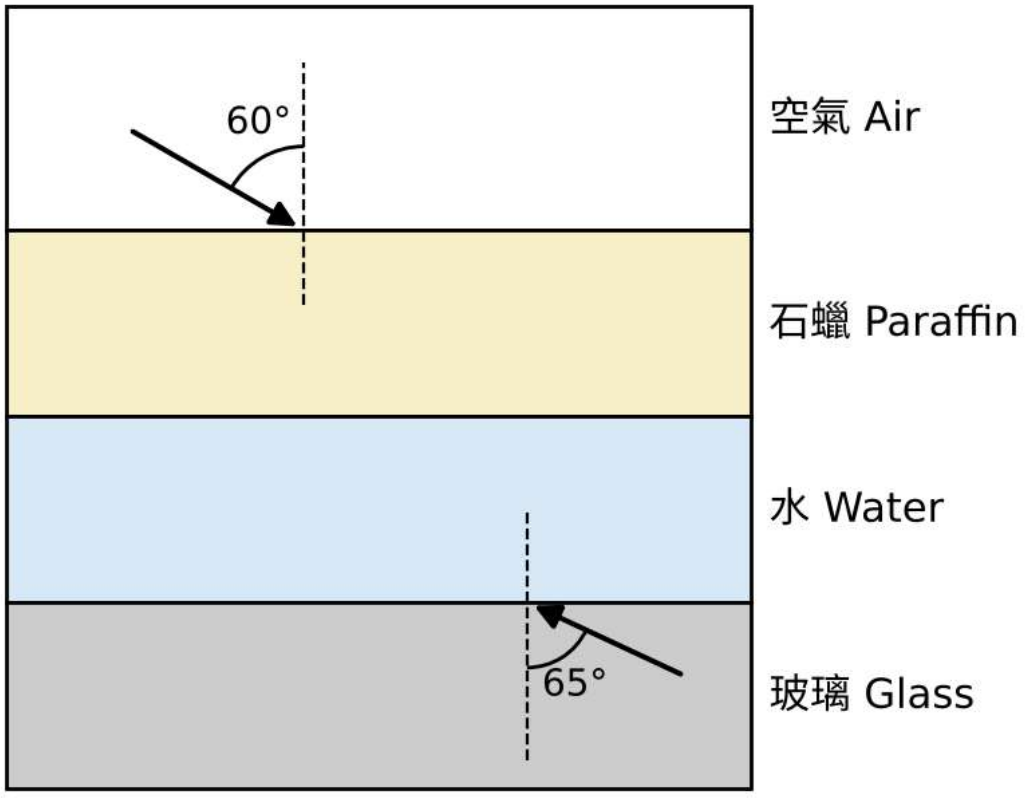


Describe how she should use the apparatus to measure the acceleration a of the bus. Your answer should include: how the apparatus is set up, the quantities to be measured, how a is calculated from the measurements (derive the relation used), and one precaution to improve accuracy. (6)

6. Read the following passage and answer the questions.

[Liquid density column]
A science museum displays a tall transparent glass cylinder. At its bottom is a thick glass base; a layer of water is poured on it, followed by a layer of liquid paraffin. Being less dense, the paraffin floats on the water, forming a flat boundary between the two liquids. The refractive indices of glass, water and paraffin are **1.52**, **1.33** and **1.47** respectively.

A guide shines a laser pointer obliquely from the air onto the top surface of the paraffin layer. The light is refracted at every boundary, but since all the boundaries are parallel, the angle θ that the ray makes with the normal in every layer satisfies one relation: $n \sin\theta$ stays the same in all layers.



(a) Using Snell's law, explain why $n \sin\theta$ stays the same in all layers, as stated in the passage. (1)

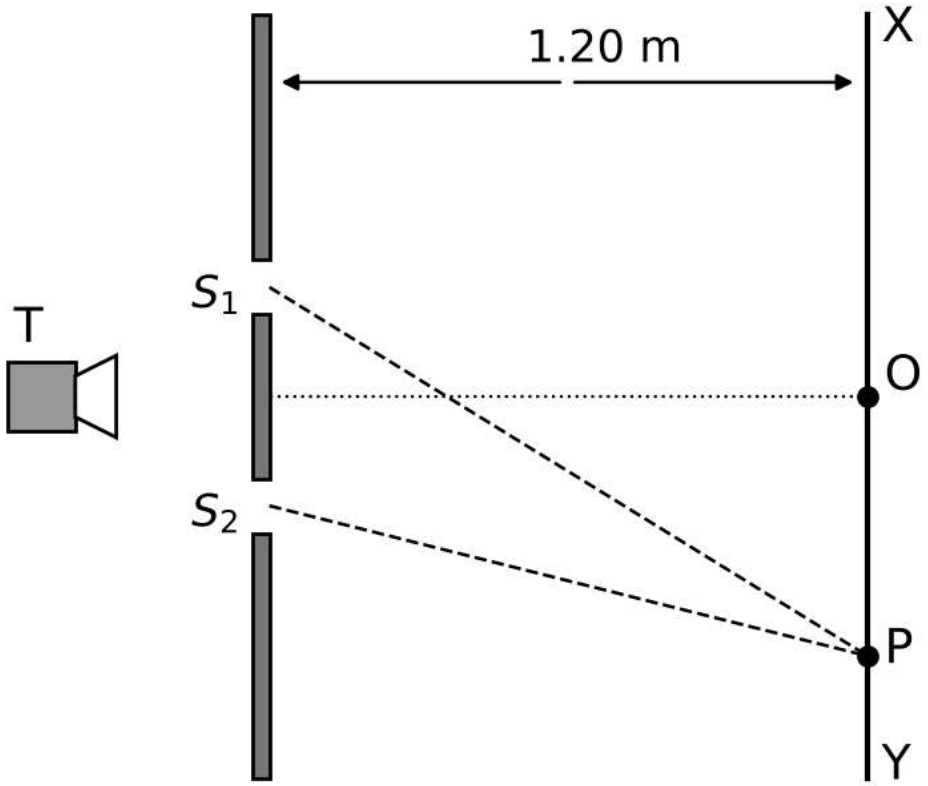
(b) The laser light strikes the top surface of the paraffin at an angle of incidence of **60°** (see Figure 1.6).

(i) Find the angles of refraction of the ray in the paraffin layer and in the water layer. (3)

(ii) The ray passes through the glass base and reaches its horizontal bottom surface, which is in contact with air outside. Determine whether the ray can emerge from the bottom surface; if so, find the angle it makes with the normal on emerging. (2)

(c) Another ray inside the glass base strikes the glass–water boundary at an angle of incidence of **65°** (see Figure 1.6). Determine whether this ray can enter the water. Support your answer with calculation. (2)

7. Figure 1.7 shows the top view of a microwave double-slit experiment. A transmitter T emits microwaves of wavelength **2.8 cm** towards a metal plate with two slits S_1 and S_2 whose centres are **6.0 cm** apart. A receiver is moved along a straight line XY parallel to the plate and **1.20 m** from it. O is on the perpendicular bisector of the slits. P is **170.2 cm** from S_1 and **166.0 cm** from S_2.



(a) As the receiver moves from X to Y, its reading rises and falls alternately. Explain this. (2)

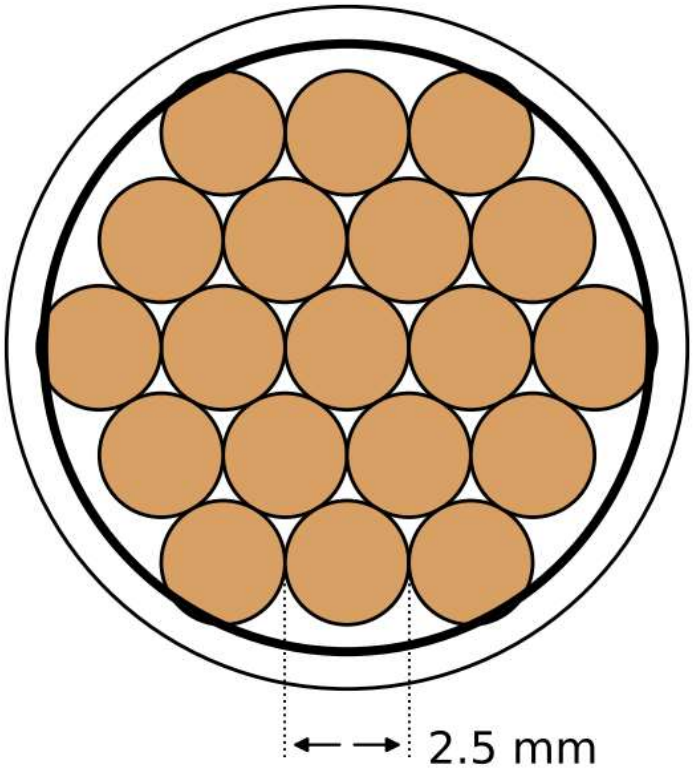
(b) Determine whether the reading at P is a maximum or a minimum, and state which maximum or minimum below O it is. (2)

(c) A student uses the formula $\Delta y = \frac{\lambda D}{a}$ to predict the distance of the first maximum from O. The measured distance is $0.63\ \text{m}$. Calculate the predicted value, and explain why the two differ noticeably. (3)

(d)(i) If line XY is extended indefinitely in both directions, what is the maximum number of maxima (including that at O) that the receiver can record? (1)

(ii) If the separation of the slit centres is reduced to **2.0 cm**, how many maxima can the receiver record along XY? Explain your answer. (2)

8. A floating solar farm transmits electrical energy to a village **8.0 km** away through a pair of underwater cables (total wire length **16 km**). Each cable consists of **19** identical copper strands twisted together, each of diameter **2.5 mm** (see Figure 1.8). The resistivity of copper is $1.7 \times 10^{-8}\ \Omega\ m$.



(a)(i) Find the resistance of one strand of length **1.00 km**. (2)

(ii) Find the resistance of a **1.00 km** length of cable. (2)

(iii) In one **1.00 km** section of cable, **2** strands are broken along the whole length. Find the percentage increase in the resistance of this section. (2)

(b) The farm sends out a power of **2.4 MW** at a voltage of **22 kV**. Assume the cables are undamaged.

(i) Find the current in the cables. (2)

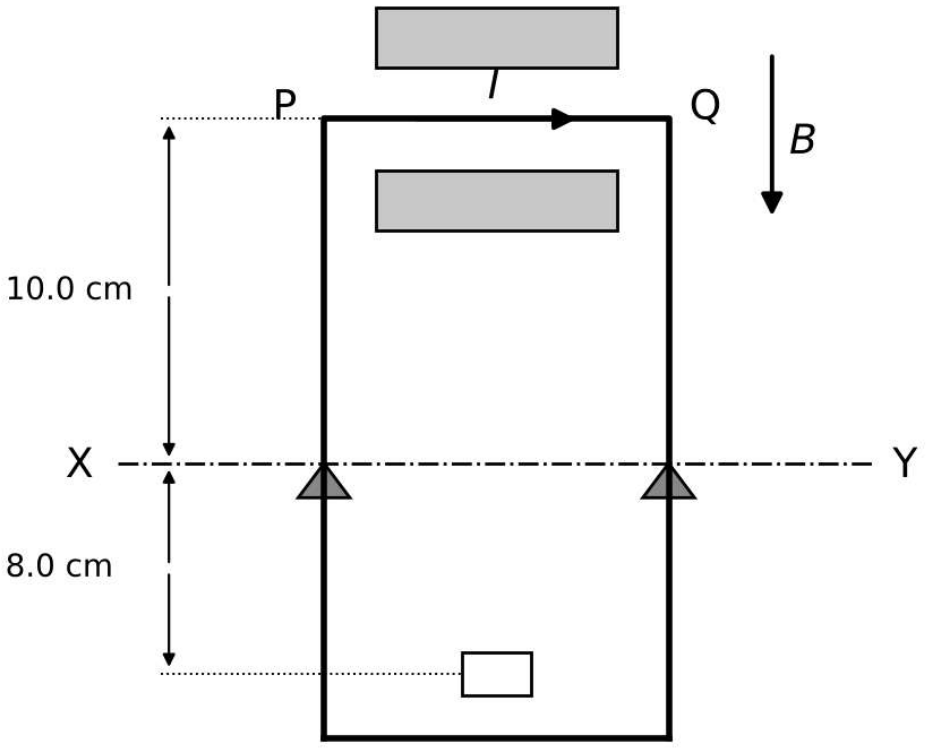
(ii) Find the power loss in the cables. (2)

(iii) If the transmission voltage is changed to **11 kV** (same power sent out), find the power loss in the cables. (1)

(iv) The village needs at least **2.30 MW**. Determine whether transmission at **11 kV** can meet the needs of the village. (2)

9. Figure 1.9 shows the top view of a "current balance". A light metal frame rests on a horizontal pivot axis XY and is free to rotate. Wire PQ of the frame, of length **5.0 cm**, is parallel to the axis, **10.0 cm** from it, and lies between a pair of magnetic poles; the magnetic field ***B*** is horizontal and perpendicular to PQ, pointing from PQ towards the axis in the figure. The other end of the frame extends to the other side of the axis. With no current the frame stays horizontal. When a current of **2.0 A** flows from P to Q as

shown, a paper rider of mass **0.15 g** must be placed on the other side of the axis, **8.0 cm** from it, for the frame to be horizontal again.



(a) Explain why, with the current on, a rider must be placed on the other side of the axis to keep the frame horizontal. Your answer should state the direction of the magnetic force on PQ. (3)

(b)(i) Find the magnitude of the magnetic force on PQ. (2)

(ii) Find the magnitude of the magnetic field B. (2)

(c)(i) If the current is reversed, can the frame be balanced simply by moving the rider to another position on the same side of the axis? Explain your answer. (1)

(ii) Suggest one way to use this balance to measure a weaker magnetic field. (1)

10. Polonium-210 is an α source. Its decay equation is:

$^{210}_{84}\text{Po} \rightarrow {}^{206}_{82}\text{Pb} + {}^{4}_{2}\text{He}$

Given masses: $^{210}_{84}\text{Po}$: **209.982874 u**; $^{206}_{82}\text{Pb}$: **205.974465 u**; $^{4}_{2}\text{He}$: **4.002603 u**.

(a) Find the energy released in each decay, in MeV. (2)

(b) A stationary polonium-210 nucleus decays.

(i) Find the kinetic energy of the α particle, in MeV. (2)

(ii) Explain why all the α particles emitted by polonium-210 have the same kinetic energy. (1)

(iii) A sealed capsule contains polonium-210 of activity 4.0×10^{11} Bq. All the α particles and recoiling lead nuclei are stopped inside the capsule. Estimate the rate at which heat is produced in the capsule. (2)

END OF PAPER